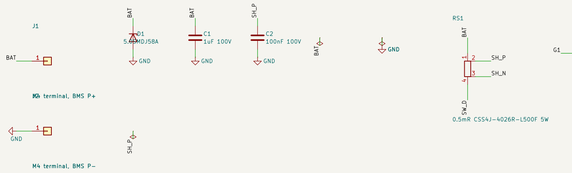
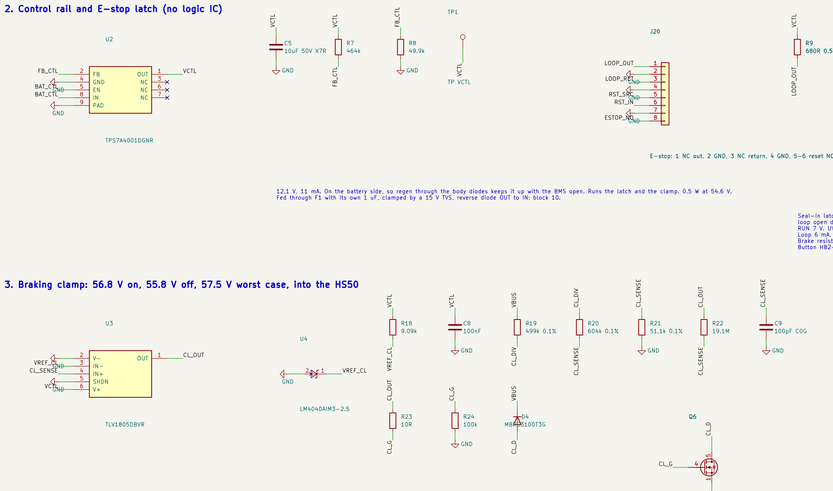
1. Battery in and E-stop bank B (LM5069-1, two FETs). Bank A, in series ahead of it, is block 7



The bank is the soft start for all the motor capacitance: BMS sees no inrush. Reset closes it. loop open drops it. No software in the path. Compute is not behind it.
R1 0.5 mOhm; limit 97–123 A (VCL 48.5–61.5 mV), breaker 160–260 A (VCB 80–130 mV). Above peak draw and JK DOC 80 A. 1.6 W at 60 A.
Start: Cdv 100 nF, gate 10–22 uA, 100–220 V/s, 0.25–0.55 s to full, Bus ceiling 9.4 mF at 2.06 A hot DC SOA line (112 W, 45 K Tj rise). Drive caps unmeasured.
Power limit 652 W at 54.6 V, 480–810 W with tolerances: RPWR = 1.3x5 x Rs x (F – 1.18 mV x VDS / Rs). SNV5452G eq 10. VSNS_6 mV over the 5 mV floor, Timer 4.7 nF: 0.22 ms typ, 0.15–0.39 ms.
Fault check: 810 W is 14.8 A at 54.6 V for up to 0.39 ms; Reviewers read about 16 A at 0.82 ms, 70 C, one FET; Tr scales as ΓD.346, so about 20.7 A at 0.39 ms: 1.4 x. Read the real curve at layout.
The 2 mA also pulls VBUS down through U1's OUT–GATE zener, against the 54 uA of its internal I MOhm SENSE–OUT path, so VBUS reads near 0 V with the banks open.
Body diodes of both banks pass regen back to the pack with the banks open. OVLO tied low: opening on overvoltage would strand regen on the bus.
LM5069-1 latches on a fault. Clear: E-stop cycle, then reset.
BAT spike: the absorber in block 10 takes the lead current when a bank opens fast (breaker trip 1.2 us). 5.0SMDJ58A is the backstop. FETs and U1 are 100 V.

2. Control rail and E-stop latch (no logic IC)



12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch and the clamp. 0.5 W at 54.6 V.
Fed through F1 with its own 1 uF, clamped by a 15 V TVS, reverse diode OUT to IN: block 10.

E-stop: 1 NC out, 2 GND, 3 NC return, 4 GND, 5–6 reset NO, 7 GND, 8 button NO

Seal-in latch, negative checked at hFE 80: stays off at power-up even with reset held, reset closes it, a 50 us contact bounce rides through.
loop open drops it in about 100 us, re-closing the loop does nothing, a reset held through a loop cycle does nothing, a new press works 5 ms after release.
Gate 6 mA, setting for a control-rated NC contact: no 12 V on the contact, a fault on either pin cannot drive RUN. UVLO 3.8 V: needs about 4.8 V on RUN_B rising. Loses VCTL, RUN drops; UVLO drops: bank opens.
Loop 6 mA, setting for a control-rated NC contact: no 12 V on the contact, without a series R. GND between loop out and return, and between return and reset: a short there opens the loop or does nothing.
Brake resistor NC thermostat: wire it in series in the loop harness, no board parts.
Button HB2–BS545, 22 mm mushroom, twist release, 1NC + 1NO, 660 V AC 10 A, control-rated only: NC in this loop, NO to the PI (block 9). RUN also enables bank A (block 7) and SPARE_SW (block 8).

3. Braking clamp: 56.8 V on, 55.8 V off, 57.5 V worst case, into the HS50

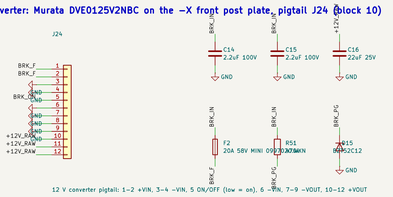
Arcol HS50 10R J on the +X gusset plate (frame 3877317), leads to J3/J4 on the board's top edge, +X half, twisted: 322 W, 5.7 A while on. MBRS3100 carries the lead and winding inductance at turn-off, so lead length sets EMI, not FET stress. 50 W on Arcol's standard heatsink, 32 g; about 1 K/W. Leg stage later: fit 2 Ohm and recheck this FET, 29 A gives about 5 W.

4. Bus capacitors

400 uF, 80 V. The drives have their own. Can is 10 x 16.5: land owed, and it only clears under the higher ceiling, away from the fans.

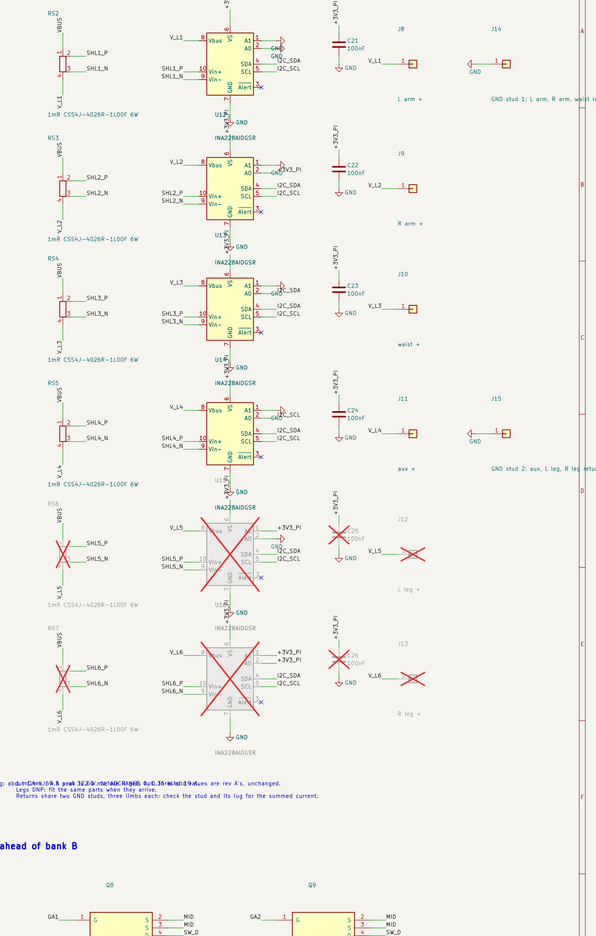


5. 12 V converter: Murata DVE0125V2NBC on the −X front post plate, pigtail J24 (block 10)

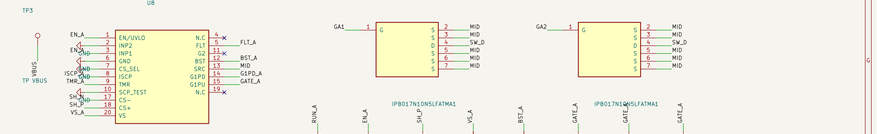


Ahead of the E-stop: motors off, compute stays up. Soft start in block 5a: the converter's negative-logic ON/OFF is held off until U7's PGD is good (block 10).
Off board on the top front rail (frame 3877317), baseplate down, 4x M3 into its threaded baseplate (50.8 x 15.24 mm, 3 mm max depth). J24 sits on the board's top edge, −X half, −Vin and −Vout both go to GND: the robot has one ground. P−.
Pigtail about 150 mm, twisted pairs, 18 AWG: 2 wires each way on the input, 3 each way on +12 V (0.1 V drop at 12 A including contacts). At the converter end: 33 uF 100 V aluminium (its ESR damps the lead, 75–150 mR) plus 1 uF across +VIN/−VIN at the pins, 300 W, 25 A, 94.6 %: about 9 W at the 140 W ceiling. Into the plate.
Input UVLO 32–36 V on, 30–34 V off, fixed on the DVE; near an empty pack under sag, compute can drop before the BMS cuts out.
INA228 addresses: L arm 0x40, R arm 0x41, waist 0x42, aux 0x43, L leg 0x44, R leg 0x45, 12 V 0x46, SPARE_SW 0x47. MID 0x48. Powered by the PI's 3.3 V: PI pull-ups on I2C.

6. Limb outputs: terminal pair, 1 mOhm shunt, INA228. Fuses inline in the pigtails.

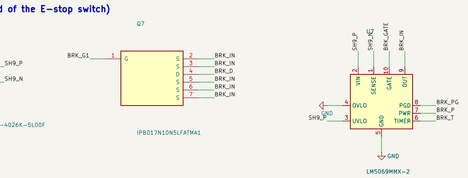


7. E-stop bank A: second series bank, own driver, ahead of bank B



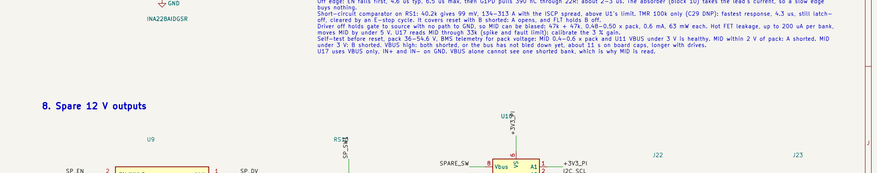
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Off board on the top front rail (frame 3877317), baseplate down, 4x M3 into its threaded baseplate (50.8 x 15.24 mm, 3 mm max depth). J24 sits on the board's top edge, −X half, −Vin and −Vout both go to GND: the robot has one ground. P−.
Pigtail about 150 mm, twisted pairs, 18 AWG: 2 wires each way on the input, 3 each way on +12 V (0.1 V drop at 12 A including contacts). At the converter end: 33 uF 100 V aluminium (its ESR damps the lead, 75–150 mR) plus 1 uF across +VIN/−VIN at the pins, 300 W, 25 A, 94.6 %: about 9 W at the 140 W ceiling. Into the plate.
Input UVLO 32–36 V on, 30–34 V off, fixed on the DVE; near an empty pack under sag, compute can drop before the BMS cuts out.
INA228 addresses: L arm 0x40, R arm 0x41, waist 0x42, aux 0x43, L leg 0x44, R leg 0x45, 12 V 0x46, SPARE_SW 0x47. MID 0x48. Powered by the PI's 3.3 V: PI pull-ups on I2C.

5a. Brick soft start (ahead of the E-stop switch)

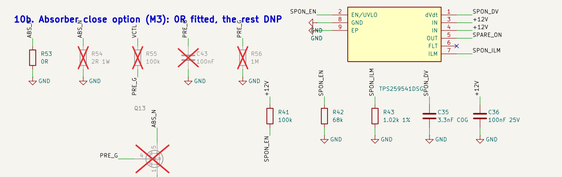


Same controller as the bank, so one part family, −2 retries on its own: a compute fault does not need a BMS cycle. OVLO on VIN. OVLO low: on whenever the pack is.
Power-limit start, go dV/dt network: 70 W limit (Rs 5 mOhm, same equation as U1), about 1.3 A at full pack, 4.4 uF/ohm, the brick's own input caps (not published) charges in under 1 ms at 20 uF.
Timer 100 nF: 47 ms typ, 3.1–8 ms. Limit 9.7–12.3 A clears the converter's worst draw, about 4.2 A at 36 V. It charges about 48 uF at the converter end plus the DVE's unpublished internal capacitance at the low power limit.
Fault check: 87 W is 1.6 A for up to 3.8 ms, against 12 A at 10 ms on the datasheet SOA. Q7 is the linear FET for that reason.
Compute is on whenever the BMS is on. Loss at 3 A: 0.05 W shunt, 0.03 W FET.

8. Spare 12 V outputs



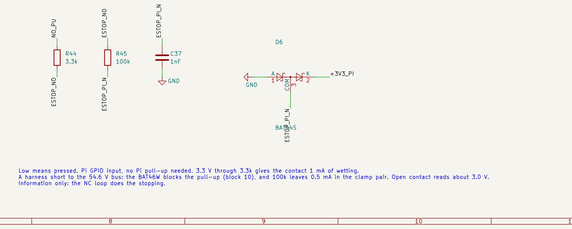
8a. SPARE_ON eFuse



Unswitched, about 2 A limit. Same part as U9: a short stops in 5 us and leaves compute up.

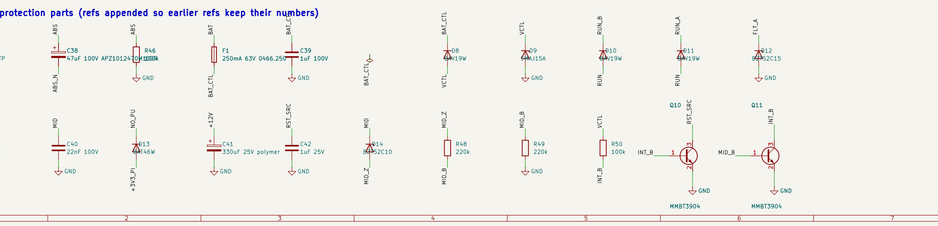
10b. Absorber close option (M3): 0R fitted, the rest DNP

9. E-stop pressed: button NO contact to the PI



Low means pressed. PI GPIO input, no PI pull-up needed. 3.3 V through 3.3k gives the contact 1 mA of wetting.
A harness short to the 54.6 V bus: the BAT46W blocks the pull-up (block 10), and 100k leaves 0.5 mA in the clamp pair. Open contact reads about 3.0 V.
Information only; the NC loop does the stopping.

10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)



11. Mechanical

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/electrical/pdb/REV_B_PARTS.md

marengo	
Sheet: /	
File: marengo_pdu_revb.kicad_sch	
Title: marengo power board rev B	
Size: A1	Date: 2026-11-06
KiCad E.D.A. 10.0.0	Rev: B-wip
	Id: 1/1